

VISVESVARAYA TECHNOLOGICAL UNIVERSITY



BELAGAVI – 590018, Karnataka

INTERNSHIP REPORT

ON

“Development of a Smart Spoon for Enhancing Taste Perception Using Electric Stimulation”

Submitted in partial fulfilment for the award of degree(21)

BACHELOR OF ENGINEERING IN AI&ML

Submitted by:

ANJALI PATEL

1BI21AI005



LOGO Ex:

Conducted at
COMPSOFT TECHNOLOGIES



BANGALORE INSTITUTE OF TECHNOLOGY

Department of AI&ML

Accredited by NBA, New Delhi

VV Pura, Bangalore -560004

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CERTIFICATE

This is to certify that the Internship titled **“Development of a Smart Spoon for Enhancing Taste Perception Using Electric Stimulation”** carried out by **Ms. ANJALI PATEL**, a bonafide student of Bangalore Institute of Technology, in partial fulfillment for the award of **Bachelor of Engineering**, in **AI&ML** under Visvesvaraya Technological University, Belagavi, during the year 2022-2023. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The project report has been approved as it satisfies the academic requirements in respect of Internship prescribed for the course Internship / Professional Practice (21CSI85)

Signature of Guide

Signature of HOD

Signature of Principal

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

D E C L A R A T I O N

I, **Anjali Patel**, final year student of AI&ML, BIT - 560 082, declare that the Internship has been successfully completed, in **Compsoft Technologies**. This report is submitted in partial fulfillment of the requirements for award of Bachelor Degree in AI&ML, during the academic year 2024-2025.

Date : 01-05-2025

:

Place : Bangalore

USN : 1BI21AI005

NAME : Anjali Patel

OFFER LETTER



Date: 6th September, 2024

Name: Anjali Patel
USN: 1BI21AI005
Intern ID: CMPSTSEP304

Dear Anjali Patel,

We are delighted to offer you a Data Science Internship position at our company, with a specific focus on medical-related projects. Your onboarding as an Intern is scheduled to begin on September 13th, 2024. We are impressed by your skills and eager to support your professional growth in this exciting and impactful field.

This internship will immerse you in the world of data-driven healthcare solutions. You'll work closely with our experienced data science team, gaining valuable insights into the application of data science techniques for medical research and innovation. Expect to tackle real-world challenges, contributing directly to projects that have the potential to improve patient outcomes and advance medical knowledge.

Highlights:

- **Potential for Extension:** Based on your performance and project needs, your internship may be extended beyond the initial timeframe.
- **Performance Recognition:** Outstanding interns may be rewarded with stipends to acknowledge their significant contributions.
- **Future Opportunities:** Exceptional performance and alignment with our company culture could lead to consideration for full-time roles upon internship completion.
- **Professional Endorsement:** A letter of recommendation will be provided upon successful completion of the internship, bolstering your future career prospects.

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. Aswath M.U., for providing usadequate facilities to undertake this Internship.

We would like to thank our Head of Dept AI&ML - Dr. Kempanna M., for providing us an opportunity to carry out Internship and for his valuable guidance and support.

We would like to thank our Keerthi Sir Software Services for guiding us during the period of internship.

We express our deep and profound gratitude to our guide, Dr. Shruthiba A., Assistant Prof, for his keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of our department for the support extended during the course of Internship.

We would like to thank the non-teaching members of our dept, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Anjali Patel
1BI21AI005

ABSTRACT

This internship report presents the work I completed during my internship at **Compsoft Technologies Private Limited**, where I was assigned a real-time data analysis project titled “**Smart Spoon Project.**” The **Smart Spoon** project aims to integrate advanced AI technologies, including computer vision, machine learning, and natural language processing, to revolutionize how users interact with food. By combining real-time **food recognition** through computer vision, **data collection** for market research, and **sentiment analysis** of user feedback, the Smart Spoon offers a personalized, data-driven experience. The system identifies various foods using Convolutional Neural Networks (CNNs) and gathers user interaction data, which is analyzed to uncover preferences, predict behaviors, and improve the recognition process. Sentiment analysis further enhances the user experience by analyzing feedback to personalize food suggestions. This project not only enhances user engagement but also provides valuable insights into food consumption patterns and user behavior, driving continuous improvements in product development and user satisfaction.

Table of Contents

Sl no	Description	Page no
1	Company Profile	9
2	About the Company	11
3	Introduction	13
4	System Analysis	15-16
5	Requirement Analysis	18
6	Design Analysis	20-25
7	Implementation	27
8	Snapshots	29-31
9	Conclusion	33
10	References	34

CHAPTER 1

COMPANY PROFILE

1. COMPANY PROFILE

A Brief History of Compsoft Technologies

Compsoft Technologies, was incorporated with a goal "To provide high quality and optimal Technological Solutions to business requirements of our clients". Every business is a different and has a unique business model and so are the technological requirements. They understand this and hence the solutions provided to these requirements are different as well. They focus on clients requirements and provide them with tailor made technological solutions. They also understand that Reach of their Product to its targeted market or the automation of the existing process into e-client and simple process are the key features that our clients desire from Technological Solution they are looking for and these are the features that we focus on while designing the solutions for their clients.

Compsoft Technologies, strive to be the front runner in creativity and innovation in software development through their well-researched expertise and establish it as an out of the box software development company in Bangalore, India. As a software development company, they translate this software development expertise into value for their customers through their professional solutions.

They understand that the best desired output can be achieved only by understanding the clients demand better. Compsoft Technologies work with their clients and help them to define their exact solution requirement. Sometimes even they wonder that they have completely redefined their solution or new application requirement during the brainstorming session, and here they position themselves as an IT solutions consulting group comprising of high caliber consultants.

They believe that Technology when used properly can help any business to scale and achieve new heights of success. It helps Improve its efficiency, profitability, reliability; to put it in one sentence "Technology helps you to Delight your Customers" and that is what we want to achieve.

CHAPTER 2

ABOUT THE COMPANY

2. ABOUT THE COMPANY



In an era defined by the relentless race for digital transformation, Compsoft Technologies stands at the forefront of empowering businesses to thrive in a globally connected, on-demand world. As industries—from healthcare, finance, and retail to energy, gaming, and transportation—scramble to build event-driven, real-time applications that capture market opportunities, Compsoft bridges the gap between cutting-edge innovation and practical implementation. Specializing in research-based, actionable data and machine learning solutions, the company pioneers unexplored domains, delivering data-supported insights that guide startups and investment firms toward strategic decision-making.

Compsoft's expertise extends beyond theoretical exploration. With a proven track record in building scalable, real-time systems and world-class consulting processes, the company equips clients with tools to operationalize innovation efficiently. By integrating advanced machine learning models with event-driven architectures, Compsoft enables businesses to unlock transformative capabilities—whether enhancing financial reporting, optimizing operational performance, or deploying AI-driven medical research platforms. Their end-to-end approach combines rigorous Theoretical Analysis, bespoke Model Development, and granular Market Research to ensure solutions are not only technologically robust but also aligned with global trends and user demands.

Driven by a mission to simplify the development and deployment of real-time applications, Compsoft delivers unparalleled cost-benefits, consistent project execution, and measurable business outcomes. Their work spans strategic goal-setting, performance management, and financial optimization, all while adhering to strict confidentiality protocols through NDAs and proprietary safeguards. For clients and interns alike, Compsoft offers a dynamic platform to shape the future of industries, merging exploratory research with scalable, real-world impact in the fast-evolving landscape of data science and digital innovation.

CHAPTER 3

INTRODUCTION

3. INTRODUCTION

Introduction to Data Science

Data science is a multidisciplinary field that combines statistics, computer science, and domain knowledge to extract meaningful insights from data. It involves collecting, cleaning, analyzing, and visualizing data to support decision-making, predict outcomes, and automate processes. With the rise of big data and machine learning, data science plays a crucial role in various industries, including healthcare, finance, and technology, enabling smarter, data-driven solutions.

Data science relies heavily on tools and techniques such as machine learning, data mining, and data visualization to uncover patterns and trends within large datasets. By using algorithms and statistical models, data scientists can forecast future events, classify information, and provide recommendations. As data continues to grow in volume and complexity, the demand for skilled data scientists increases, making it one of the most valuable and impactful fields in today's digital world.

Problem Statement

Traditional eating utensils offer no interactive features or intelligent feedback, limiting the potential for personalized dietary support and user engagement. There is a need for an innovative solution that can automatically recognize food items, understand user preferences, and analyze feedback to offer a more informed, personalized, and data-driven eating experience. Additionally, current food tracking methods lack real-time capabilities and often rely on manual input, making them inefficient and prone to user error.

CHAPTER 4

SYSTEM ANALYSIS

4. SYSTEM ANALYSIS

1. Existing System

Several existing systems and research efforts aim to enhance dietary management and taste modulation, especially for individuals with health conditions like hypertension and kidney disease. Traditional solutions include low-sodium alternatives, diet-tracking mobile apps, and manual nutrition consultation, which often require active user participation and lack real-time feedback or personalization.

A notable innovation in this space is the "Electric Taste Simulation" approach, where weak electrical currents are applied to the tongue to alter or enhance taste sensations without altering the actual food content. Some prototypes and research studies, such as electric chopsticks and taste-altering utensils, have demonstrated the potential to simulate salty, sour, or umami flavors using electrical stimulation. However, these systems are typically not data-driven and lack integration with personalized health profiles or AI-based recommendation engines.

Moreover, while wearable health devices (like fitness trackers) can monitor vital signs and dietary intake, they do not actively assist in modifying taste experience. There remains a gap in integrated smart devices that combine real-time data analysis, health profiling, taste modulation, and AI-powered recommendations—all of which your Smart Spoon project aims to address.

2. Proposed System

The proposed system is an AI-powered Smart Spoon designed to enhance taste perception using weak electrical currents, specifically tailored for individuals with low-sodium dietary needs. This system integrates multiple advanced technologies—such as computer vision, machine learning, and natural language processing—to provide a personalized and data-driven taste enhancement experience.

A. Computer Vision & Food Recognition

The system uses Convolutional Neural Networks (CNNs) to recognize different food types through image input. By accurately identifying the food item, the model predicts its salt content and dynamically adjusts the electrical stimulation levels delivered by the Smart Spoon to simulate an

enhanced salty taste. This allows users to enjoy the sensation of salt without actual sodium intake. Additionally, the system will incorporate automated dietary recommendation algorithms to guide users toward healthier meal choices based on their recognized food patterns.

B. Data Collection & Analysis for Market Research

To align the device with real-world needs, the system will include AI-driven surveys and user interaction tracking to collect consumer preferences. This data will be analyzed to identify trends, usage patterns, and areas for improvement. Insights derived from this analysis will help refine the taste simulation technology and evaluate the impact of the device on sodium-restricted diets, offering measurable health benefits over time.

C. Sentiment Analysis & User Behavior Prediction

The system will apply Natural Language Processing (NLP) techniques to process and analyze user feedback, reviews, and usage logs. This enables the identification of emotional responses, satisfaction levels, and areas of discomfort or confusion. Using this information, the system will continuously adapt and personalize user experiences, while also predicting future improvements or feature updates based on sentiment trends and behavioral patterns.

3. Objective of the System

Develop a smart spoon that simulates salty and umami taste perception using electric stimulation, without requiring actual sodium intake, thereby improving the eating experience for patients with dietary restrictions.

CHAPTER 5

REQUIREMENT ANALYSIS

5. REQUIREMENT ANALYSIS

Hardware Requirement Specification

- Intel i7
- 16GB RAM
- 256 GB SSD

Software Requirement Specification

- Jupyter Notebook
- Keras
- NumPy
- Matplotlib
- Transformers
- Pandas

CHAPTER 6

DESIGN ANALYSIS

6. DESIGN & ANALYSIS

1. System Architecture

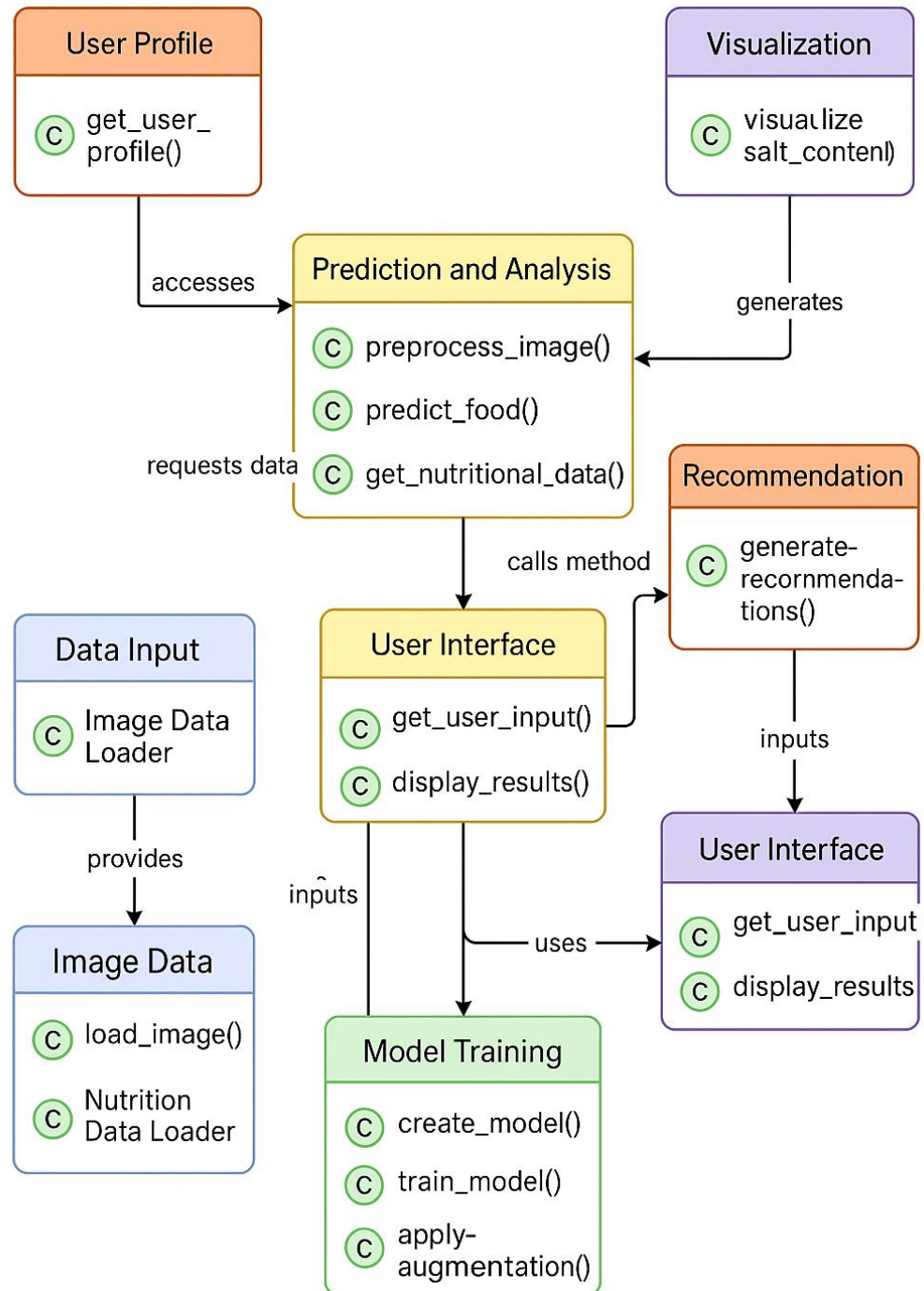


Fig 6.1 System Architecture of Nutritional Prediction & Food Recognition

Module Description

1. User Profile Module

Function:

This module retrieves the personal information and dietary preferences of the user.

Purpose:

Helps personalize predictions and recommendations.

Ensures the system tailors output to health conditions like hypertension or kidney disease.

Main Method:

`get_user_profile()`

2. Prediction and Analysis Module

Function:

The core of the system that processes input data, performs food recognition, and fetches nutritional details.

Submodules & Methods:

`preprocess_image()` – Prepares image data for analysis (e.g., resizing, normalization).

`predict_food()` – Uses a trained CNN to classify the food in the image.

`get_nutritional_data()` – Fetches nutritional content (especially sodium level) of the identified food.

Purpose:

Enables accurate prediction of food type and its salt content.

3. Recommendation Module

Function:

Generates personalized dietary suggestions or alternatives based on nutritional analysis and user profile.

Main Method:

`generate_recommendations()`

Purpose:

Helps users make healthier dietary choices aligned with their restrictions.

4. Visualization Module

Function:

Presents nutritional data (especially salt content) in an easy-to-understand format, such as charts or graphs.

Main Method:

visualize_salt_content()

Purpose:

Improves user awareness of food content and supports informed decision-making.

5. User Interface Module

Function:

Handles all user interactions with the system, including input submission and result display.

Main Methods:

get_user_input() – Accepts inputs like images or preferences.

display_results() – Shows prediction, recommendations, and visualizations to the user.

Note: Appears twice in the diagram—this reflects multiple interaction points, not duplication.

6. Data Input Module

Function:

Provides essential input data to the system, particularly image and nutritional data.

Components:

Image Data Loader – Loads images captured by the user.

Nutrition Data Loader – Loads nutritional database used for reference.

Purpose:

Acts as a data pipeline feeding raw inputs into the system.

7. Image Data Module

Function:

Directly loads and organizes image data for processing.

Main Method:

load_image()

Purpose:

Prepares food image input for the Prediction module.

8. Model Training Module

Function:

Builds and trains the machine learning models used for food recognition and data prediction.

Main Methods:

create_model() – Builds the architecture (e.g., CNN).

train_model() – Trains the model using dataset.

apply_augmentation() – Enhances training with data augmentation techniques (rotation, flipping, etc.).

Purpose:

Ensures high accuracy of prediction by learning from diverse data.

2. System Architecture

System Architecture of Sentiment Analysis & User Behavior Prediction

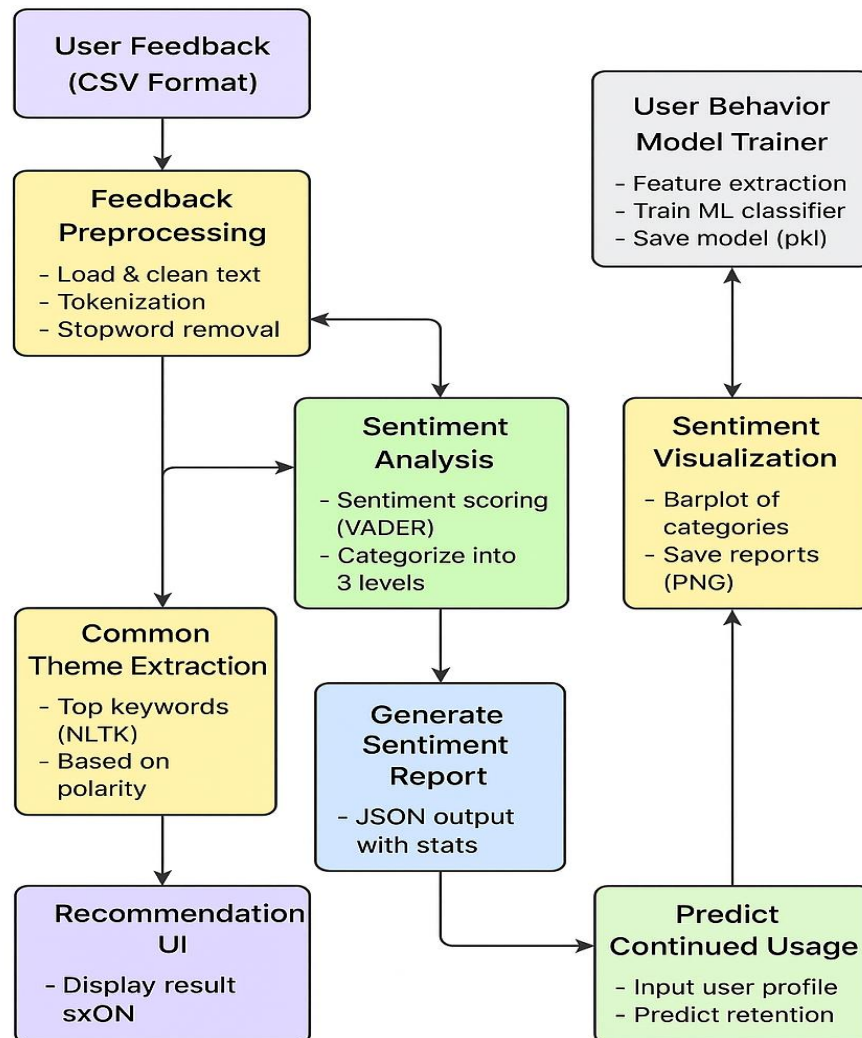


Fig 6.2 System Architecture of Sentiment Analysis & User Behavior Prediction

Module Description

1. User Feedback Analysis Module

Function:

Analyzes textual feedback from users to understand satisfaction levels and extract common sentiment trends.

Main Method:

`analyze_user_feedback(feedback_data_path)`

Purpose:

Improves personalization by understanding user opinions and detecting areas for enhancement through sentiment scoring and keyword extraction.

2. Sentiment Scoring Module

Function:

Computes sentiment polarity scores using a pre-trained sentiment analyzer to quantify user emotions.

Main Method:

`sentiment_analyzer.polarity_scores(text)`

Purpose:

Quantifies feedback into measurable sentiment values to classify them as Positive, Neutral, or Negative.

3. Keyword Extraction Module

Function:

Extracts the most frequent and relevant words from positive and negative feedback to determine key user concerns and preferences.

Main Method:

`extract_keywords(text_series)`

Purpose:

Identifies recurring themes in feedback that can guide improvements in product features and user experience.

4. Sentiment Visualization Module

Function:

Creates bar charts to visualize sentiment distribution and feedback themes across users.

Main Method:

`visualize_sentiment_distribution(feedback_data)`

Purpose:

Presents sentiment insights in a graphical format to support decision-making and reporting.

5.Sentiment Report Generation Module

Function:

Compiles sentiment metrics and extracted themes into a structured report format for future reference and analytics.

Main Method:

`generate_sentiment_report()`

Purpose:

Provides a summarized and shareable JSON report of overall user sentiment and behavioral patterns.

6.User Behavior Training Module

Function:

Trains a machine learning model to predict whether users will continue using the Smart Spoon based on behavior and demographic features.

Main Method:

`train_user_behavior_model(user_data_path)`

Purpose:

Builds a predictive model to identify long-term user engagement and potential drop-off risks.

7.User Retention Prediction Module

Function:

Predicts if a new user will continue using the device using the trained machine learning model.

Main Method:

`predict_user_retention(user_data)`

Purpose:

Enables proactive measures by forecasting user retention likelihood based on input profile data.

CHAPTER 7

IMPLEMENTATION

7. IMPLEMENTATION

Implementation is the stage where the theoretical design is turned into a working system. The most crucial stage in achieving a new successful system and in giving confidence on the new system for the users that it will work efficiently and effectively.

The system can be implemented only after thorough testing is done and if it is found to work according to the specification. It involves careful planning, investigation of the current system and its constraints on implementation, design of methods to achieve the change over and an evaluation of change over methods as a part from planning.

Two major tasks of preparing the implementation are education and training of the users and testing of the system. The more complex the system being implemented, the more involved will be the system analysis and design effort required just for implementation.

The implementation phase comprises of several activities. The required hardware and software acquisition is carried out. The system may require some software to be developed. For this, programs are written and tested. The user then changes over to his new fully tested system and the old system is discontinued.

TESTING

The testing phase is an important part of software development. It is the Information zed system will help in automate process of finding errors and missing operations and also a complete verification to determine whether the objectives are met and the user requirements are satisfied. Software testing is carried out in three steps:

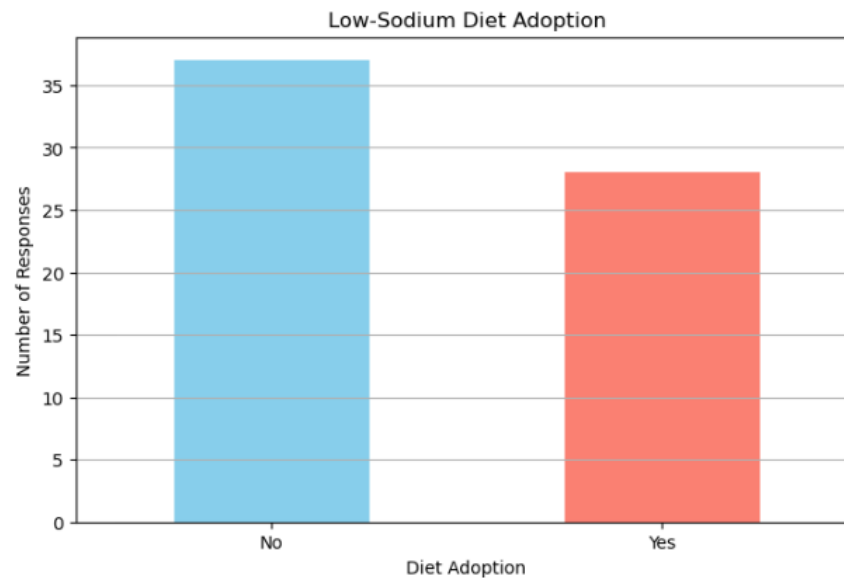
1. The first includes unit testing, where in each module is tested to provide its correctness, validity and also determine any missing operations and to verify whether the objectives have been met. Errors are noted down and corrected immediately.
2. Unit testing is the important and major part of the project. So errors are rectified easily in particular module and program clarity is increased. In this project entire system is divided into several modules and is developed individually. So unit testing is conducted to individual modules.
3. The second step includes Integration testing. It need not be the case, the software whose modules when run individually and showing perfect results, will also show perfect results when run as a whole.

CHAPTER 8

SNAPSHOTS

8. SNAPSHOTS

Low-sodium diet adoption:
Do you follow a low-sodium diet?
No 37
Yes 28
Name: count, dtype: int64
Percentage following low-sodium diet: 43.1%



Medical conditions among low-sodium dieters:
If yes, is it due to a medical condition? (Type no incase of no)
Hypertension 11
No 7
Thyroid 2
Kidney Disease 2
Blood pressure 1
I am aware 1
Cardio problem 1
Yes 1
Name: count, dtype: int64

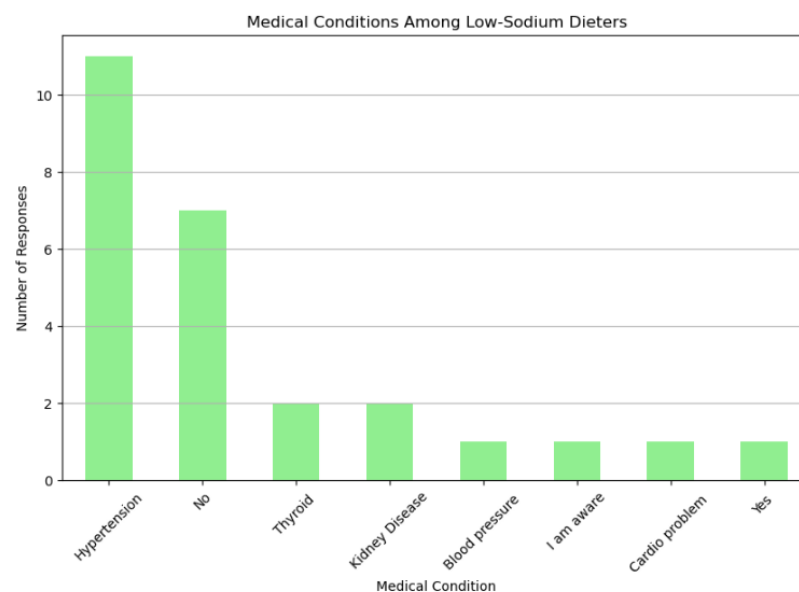
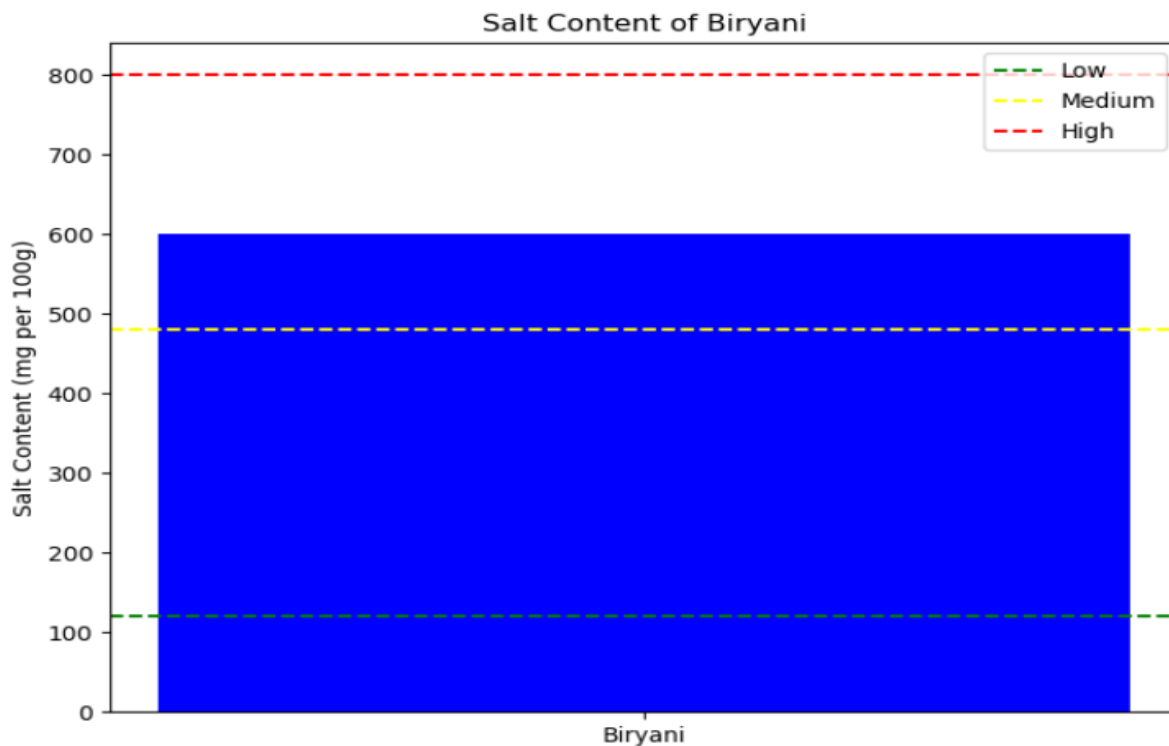


Fig 8.1 Data Collection & Analysis for Market Research

Loaded existing model
Enter user ID (1-25) or press Enter for default user ID 1: 10
1/1 2s 2s/step

===== FOOD ANALYSIS RESULTS =====
Recognized Food: Biryani (Confidence: 0.41)
Category: Main Course | Region: Hyderabadi
Main Ingredients: Rice/Meat/Spices
Salt Content: 600 mg per 100g
Typical Portion: 250g containing 1500 mg salt
Recommended Stimulation Level: 9.40/10



Dietary Recommendations:

1. Biryani contains approximately 600 mg of salt per 100g.
2. A typical portion (250g) contains 1500 mg of salt.
3. This represents approximately 100.0% of your recommended daily salt limit (1500 mg).
4. Consider pairing this dish with fresh vegetables or a simple salad to balance salt intake.
5. Recommended Smart Spoon stimulation level: 9.40/10

Fig. 8.2 Computer Vision & Food Recognition

Please enter your feedback about the Smart Spoon (or type 'exit' to quit): Bad
Predicted Satisfaction: 1

Analysis Results:

Sentiment: Negative (Score: 0.00)

Predicted Satisfaction Level: 1

Improvement Suggestions:

1. Focus on core functionality improvements for better user satisfaction.
2. Consider a user experience redesign based on customer feedback.
3. Implement quality control measures to address potential issues.

Please enter your feedback about the Smart Spoon (or type 'exit' to quit): Good

Predicted Satisfaction: 4

Analysis Results:

Sentiment: Positive (Score: 1.00)

Predicted Satisfaction Level: 4

Improvement Suggestions:

1. Enhance features that receive the most positive feedback.
2. Consider expanding product line with complementary offerings.
3. Develop ambassador programs to leverage satisfied customers.



Fig 8.3 Sentiment Analysis & User Behavior Prediction

CHAPTER 9

CONCLUSION

9. CONCLUSION

The package was designed in such a way that future modifications can be done easily. The following conclusions can be deduced from the development of the project:

- ❖ Automation of the entire system improves the efficiency
- ❖ It provides a friendly graphical user interface which proves to be better when compared to the existing system.
- ❖ It gives appropriate access to the authorized users depending on their permissions.
- ❖ It effectively overcomes the delay in communications.
- ❖ Updating of information becomes so easier
- ❖ System security, data security and reliability are the striking features.
- ❖ The System has adequate scope for modification in future if it is necessary.

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